Response-aware path compression

Sharp planar distortion and exact Euler storage

Davin

14 October 2026

Abstract

Path compression stores a driving signal while controlling its effect on differential equations. This discussion draft compares worst-case terminal error over jointly Frobenius-bounded linear systems with a weighted Hilbert–Schmidt signature-error certificate.

For planar bounded-variation paths, storing displacement and area on each interval and decoding by matrix exponentials gives a response-to-certificate ratio tending to $2\sqrt{2}/9$ under small path scaling. This requires a fixed partition and nonzero cubic discrepancy; convergence is uniform in response dimension $n \geq 2$. Explicit scale and perturbation bounds certify finite compression decisions. Chains of adjacent area-only blocks admit a strict gap at every positive scale.

For Euler decoding, we derive exact bounded-variation minimax error under path-dependent, coefficient-independent partitions. Replacing a length envelope by this constraint can change optimal truncation depth and interval count. Finite storage costs decrease in examples, while leading asymptotics agree.

1 Compression and response error

A compressed path representation stores interval summaries from which a system's response can be computed. In (ε, r)-representation framework [1], terminal error must be at most ε for every linear system with coefficient budget r. We compare this requirement with a tensor-error certificate to determine whether response accuracy permits cheaper storage.

1.1 Stored data and the decoder

Let $X : [0,1] \to \mathbb{R}^d$ be continuous with finite Euclidean variation

$$\|X\|_{1\text{-var}} = \sup_{\tau} \sum_{[u,v]\in\tau} \|X_v - X_u\|_2 ,$$

where τ ranges over finite partitions of $[0,1]$. Here $\|\cdot\|_2$ is Euclidean norm; BV abbreviates bounded variation. For $\pi = (0 = t_0 < \cdots < t_m = 1)$, write $X_i = X|_{[t_{i-1},t_i]}$. Its signature collects iterated integrals:

$$S(X)_0 = 1, \qquad S(X)_j = \int_{0<u_1<\cdots<u_j<1} dX_{u_1} \otimes \cdots \otimes dX_{u_j}, \quad j \geq 1.$$

For $S(X_i)$, use integration times in $[t_{i-1}, t_i]$. Tensor multiplication concatenates words. For $U_0 = 0$, define $e^U_\otimes = \exp_\otimes U = \sum_{q\geq 0} U^{\otimes q}/q!$, with $U^{\otimes 0} = 1$; $\log_\otimes$ is its formal inverse on series with constant term one.

Store one truncated log-signature per interval. Degree-two truncation retains displacement and signed area in the plane:

$$\ell_i^{\leq 2} = \ell_{i,1} + \ell_{i,2}, \qquad \ell_{i,j} = (\log_\otimes S(X_i))_j.$$

Indices i, j denote interval and tensor degree. Exponentiating stored logs and multiplying in time order gives

$$F = S(X), \qquad G_\pi = \prod_{i=1}^{m} \exp_\otimes(\ell_i^{\leq 2}).$$

Thus F is exact signature and G_π is decoded series; only $(\ell_i^{\leq 2})_{i=1}^m$ is stored.

Fix a tuple $A = (A_1, \ldots, A_d)$ of real $n \times n$ matrices, with d driver channels and n response states. Its row-state system is

$$dY = \sum_{i=1}^{d} Y A_i \, dX^i, \qquad Y_0 = I_n.$$

Here I_n is identity matrix. For standard driver basis $(e_i)_{i=1}^d$, define $R_A(e_{i_1} \otimes \cdots \otimes e_{i_j}) = A_{i_1} \cdots A_{i_j}$, extended linearly over convergent series, with empty word mapped to I_n. Brackets denote commutators: $[U, V] = U \otimes V - V \otimes U$ for tensors, and $[A, B] = AB - BA$ for matrices. Exact and decoded terminal responses are

$$\Phi^A(X) = R_A(F), \qquad \widehat{\Phi}_\pi^A(X) = R_A(G_\pi) = \prod_{i=1}^{m} \exp\big(R_A(\ell_i^{\leq 2})\big).$$

Lie polynomials are linear combinations of nested commutators. Degree-two log-ODE decoding evaluates each stored Lie polynomial, exponentiates, then composes. One tuple A acts on every interval. Transposing tuples gives column-state convention with unchanged norms.

$$X \xrightarrow{\text{solve exactly}} \Phi^A(X),$$

$$X \xrightarrow{\text{split } \pi} \underbrace{(\ell_i^{\leq 2})_{i=1}^m}_{\text{finite stored data}} \xrightarrow{\text{exponentiate and compose}} \widehat{\Phi}_\pi^A(X).$$

1.2 True response error versus a tensor certificate

Impose coefficient budget $|A|^2 = \sum_i \|A_i\|_F^2 \leq r^2$, $r > 0$, where $\|A_i\|_F^2 = \operatorname{tr}(A_i^\top A_i)$. For $F_0 = G_0 = 1$, true response error is

$$\Delta_{r,n}(F, G) = \sup_{|A| \leq r} \|R_A(F) - R_A(G)\|_{\mathrm{op}}. \tag{1}$$

Here $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2 = 1} \|Mv\|_2$, so this bounds error for every unit initial state. Degree-two decoding uses $G = G_\pi$.

For degree components F_j, G_j, declare standard word tensors orthonormal. Hilbert–Schmidt (HS) norm is Euclidean norm of word coefficients. Define

$$\mathcal{B}_r(F, G) = \sum_{j \geq 1} r^j \, \|F_j - G_j\|_{\mathrm{HS}}.$$

Degreewise Cauchy–Schwarz gives $\Delta_{r,n} \leq \mathcal{B}_r$; the series converge for representations considered here. Tolerance $\varepsilon > 0$ is met when $\Delta_{r,n} \leq \varepsilon$. Certificate condition $\mathcal{B}_r \leq \varepsilon$ is sufficient and can reject response-feasible representations.

For straight line $X_u = ue_1$ compared with zero signal, both errors equal $e^r - 1$: choose $A_1 = rQ$ for a rank-one orthogonal projection Q, and other matrices zero. Sections 2–3 use Lie structure under log-ODE decoding; Section 4 treats Euler decoding.

2 Planar log-ODE error: the first discarded term

Fix $d = 2$ and partition π. Exact and decoded interval series agree through degree two, so their first possible discrepancies add:

$$D_3 := (F - G_\pi)_3 = \sum_i \ell_{i,3}.$$

Dropped cubic logs determine leading error when $D_3 \neq 0$.

Scale X to sX, $s > 0$, keeping r fixed. Tensor dilation is $(\delta_s F)_j = s^j F_j$. With $t = sr$, write

$$\Delta_n(t) = \Delta_{1,n}(\delta_t F, \delta_t G_\pi), \qquad \mathcal{B}(t) = \mathcal{B}_1(\delta_t F, \delta_t G_\pi).$$

These equal errors for sX at budget r.

Theorem 1. *If $D_3 \neq 0$, then for every $n \geq 2$,*

$$\frac{\Delta_n(t)}{\mathcal{B}(t)} \longrightarrow \rho := \frac{2\sqrt{2}}{9} \quad (t \downarrow 0),$$

uniformly in n. Explicit positive scale and BV perturbation radii ensure $\Delta_n(t) < \mathcal{B}(t)$; formulas appear in Appendix A.

Why this constant? HS norm counts word coefficients independently. Matrix products share one budget-constrained tuple, and cubic Lie terms evaluate to nested commutators. A driver rotation reduces any nonzero planar cubic direction to a multiple of $[e_1, [e_1, e_2]]$. Its coefficients $1, -2, 1$ give HS norm $\sqrt{6}$. Böttcher–Wenzel applied twice [2] and joint unit budget give matrix-response maximum $4/(3\sqrt{3})$, attained in dimension two. Hence

$$\frac{4/(3\sqrt{3})}{\sqrt{6}} = \frac{2\sqrt{2}}{9}.$$

Thus leading certificate error exceeds response error by factor $1/\rho \approx 3.18$. Appendix A proves attainment and bounds higher-degree tails.

A finite compression decision. For corner $0 \to e_1 \to e_1 + e_2$, store displacement and area in one interval. At unit coefficient budget,

$$\Phi^A(tX) = e^{tA_1} e^{tA_2}, \qquad \widehat{\Phi}^A_\pi(tX) = e^{t(A_1+A_2)+(t^2/2)[A_1,A_2]}.$$

At $t = 1/20$, rational enclosures prove, for every $n \geq 2$,

$$\Delta_n(t) < 1.934 \times 10^{-5} < 2 \times 10^{-5} < 3.608 \times 10^{-5} < \mathcal{B}(t). \tag{2}$$

Tolerance 2×10^{-5} accepts this representation under response bound and rejects it under full HS certificate. Appendix F supplies rational certificates.

A two-interval square loop can have $D_3 = 0$ and nonzero quartic error; partition refinement can make D_3 vanish. Mesh-uniform bounds and extension to rough inputs with roughness exponent $2 < p < 3$ remain open.

3 Homogeneous merges at every positive scale

Merging stored area blocks replaces their ordered product by an exponential of their sum. Area is preserved; noncommuting matrix responses can change.

For $\ell \geq 3$ blocks in $d = \ell + 1$ channels, let $U_i = [e_i, e_{i+1}]$, $1 \leq i \leq \ell$, and compare

$$F_\lambda = \prod_{i=1}^{\ell} e_\otimes^{\lambda U_i}, \qquad G_\lambda = e_\otimes^{\lambda \sum_i U_i}, \qquad \lambda > 0.$$

Here U_i represents area in coordinate plane $(i, i+1)$, with amplitude λ. At unit tuple budget, write $\Delta_n(\lambda) = \Delta_{1,n}(F_\lambda, G_\lambda)$ and $\mathcal{B}(\lambda) = \mathcal{B}_1(F_\lambda, G_\lambda)$.

Theorem 2. *For every $\ell \geq 3$, every response dimension $n \geq 1$ and $\lambda > 0$,*

$$\Delta_n(\lambda) < \frac{\lambda^2}{\sqrt{2}} e^{\sqrt{2}\lambda} < \frac{e^{\sqrt{6}\lambda} - 1 - \sqrt{6}\lambda}{\sqrt{3}} \leq \mathcal{B}(\lambda). \tag{3}$$

The separation bounds are independent of chain length.

Response bounds permit an ℓ-to-one merge for tolerances rejected by full certificate. Fixed generic depth-two layout reduces storage by factor ℓ. For truncated BV loops, total error also includes initial decoding error, by triangle inequality. Pure area rough blocks have exact fine series. Appendix B proves all-scale bounds by coefficient comparison.

4 Exact Euler minimax and storage

Euler decoding evaluates a truncated signature polynomial, reconstructed from stored log-signature coordinates. Choose depth N and interval count m to minimize storage subject to response tolerance.

For positive integers N, m, put $P_N(x) = \sum_{j=0}^{N} x^j/j!$. Write $R_{A,j}$ for the restriction of R_A to degree j. The decoded response is

$$E^A_{N,\pi}(X) = \prod_{i=1}^{m} \left[I_n + \sum_{j=1}^{N} R_{A,j}(S(X_i)_j) \right].$$

For each path, choose one partition controlling every admissible tuple.

Theorem 3. *For $N, m, d, n \geq 1$, $r > 0$, and continuous paths $X : [0,1] \to \mathbb{R}^d$ with $\|X\|_{1\text{-}var} \leq 1$,*

$$\sup_X \inf_{\pi:\, m \text{ intervals}} \sup_{|A| \leq r} \left\| \Phi^A(X) - E^A_{N,\pi}(X) \right\|_{\mathrm{op}} = \mathcal{E}(N, m, r) := e^r - P_N(r/m)^m. \tag{4}$$

Partitions are path-dependent and coefficient-independent. Equal-variation partitions give upper bound; a straight line with rank-one development attains equality.

The (ε, r)-representation paper gives this scalar expression in (1.13). Appendix C proves universal upper bound and partition minimization. Their length envelope is

$$B(N, m, r) = \frac{e^r r^{N+1}}{(N+1)! m^N}.$$

Fix Euler decoder and generic coordinate layout. Let $\Lambda_d(N)$ count independent log-signature coordinates through degree N in d channels. Storage is $m\Lambda_d(N)$ scalars. Appendix D gives dimension formula and storage asymptotics.

Finite integer movement. For $d = 2, r = 1, \varepsilon = 10^{-2}$, the two optima are

Constraint	N	m	stored scalars
Exact BV minimax	4	1	8
Length envelope	5	1	14

Appendix F certifies global minimality with rational arithmetic. On the retained grid $d \in \{2, 3, 5\}$, $r \in \{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$ and $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-4}, 10^{-8}, 10^{-12}\}$, optimum sets change in 85 of 135 cases; 84 have strictly lower storage, with largest saving 42.9%. Grid searches use high-precision boundary refinement without interval certification.

Storage limits. At fixed $d \geq 2$, the ratio of minimum costs under $\mathcal{E} \leq \varepsilon$ and $B \leq \varepsilon$ tends to one in either limit: $\varepsilon \downarrow 0$ at fixed $r > 0$, or $r \to \infty$ at fixed $\varepsilon > 0$. Appendix D proves this and gives depth, interval and cost asymptotics.

For pure Engel directions, obtained by repeated bracketing of two drivers, HS-certified Euler storage overhead grows with amplitude. A single pure log-signature block is exact under log-ODE decoding. Appendix E states and proves the storage bracket.

5 Prior work and boundaries

Development separation through polynomial identities appears in Chevyrev–Lyons, §4.2 and Theorem 4.8 [3]. Planar degree-six matrix identities [4] obstruct separation by dimension-two responses. Spectral Lie-polynomial evaluations and low-degree calculations appear in Boedihardjo–Geng–Souris [5] under different norm budgets; exact pure-path tail asymptotics appear in Li–Gao [6]. The Engel gauge below is a Böttcher–Wenzel corollary [2]. Product-formula bounds appear in [7]; fixed-system adaptive log-ODE analysis in [8]. The (ε, r)-representation paper identifies first discarded local log-signature error after (2.10).

Novelty of constrained response/HS comparison and finite-scale consequences requires confirmation. General rough inputs with $2 < p < 3$, mesh-uniform nondegeneracy, exact finite-scale suprema and general cubic-zero behavior remain open. General rough-path asymptotics and empirical averaged-error figures in the (ε, r)-representation paper are outside these results' scope.

A Planar gauge, tails and stability

Write $S_j(X) = S(X)_j$ and E_{ij} for a matrix unit with entry one at (i,j) and zero elsewhere. For homogeneous D of degree j, write $\gamma_{j,n}(D) = \sup_{|A|\le 1} \|R_A(D)\|_{\mathrm{op}}$. Let $Z = [e_1, e_2]$. Every planar cubic Lie element is $D = [w, Z]$ with $w \in \mathbb{R}^2$, and $\|D\|_{\mathrm{HS}} = \sqrt{6}\,\|w\|_2$. Proper driver rotations preserve tuple ball and fix Z, reducing to $D = \|w\|_2\,[e_1, Z]$. Böttcher–Wenzel and scalar maximization give

$$\|[A_1,[A_1,A_2]]\|_{\mathrm{op}} \le 2\,\|A_1\|_F^2\,\|A_2\|_F \le \frac{4}{3\sqrt{3}}.$$

Equality holds at

$$A_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad A_2 = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

whose cubic response is $4E_{12}/(3\sqrt{3})$. Embedding these matrices proves equality for $n \ge 2$; hence $\gamma_{3,n}(D) = \rho\,\|D\|_{\mathrm{HS}}$. One commutator step similarly gives $\gamma_{2,n}(\alpha Z) = |\alpha|/\sqrt{2} = \|\alpha Z\|_{\mathrm{HS}}/2$. It is attained at $A_1 = \mathrm{diag}(1,-1)/2$ and $A_2 = E_{12}/\sqrt{2}$.

Explicit tails and radius. Let $L = \|X\|_{1\text{-var}}$ and put

$$h = \|D_3\|_{\mathrm{HS}}, \qquad a = \sum_i \|\ell_{i,1}\|_2, \qquad b = \sum_i \|\ell_{i,2}\|_{\mathrm{HS}}.$$

Define scalar majorant tail from degree four:

$$T_4(a,b;t) = e^{at+bt^2} - 1 - at - (a^2/2 + b)t^2 - (a^3/6 + ab)t^3.$$

Its coefficients are nonnegative. Define $Q_R(t) = T_4(L,0;t) + T_4(a,b/2;t)$ and $Q_H(t) = T_4(L,0;t) + T_4(a,b;t)$. Then

$$|\Delta_n(t) - \rho h t^3| \le Q_R(t), \qquad ht^3 \le \mathcal{B}(t) \le ht^3 + Q_H(t). \tag{5}$$

For $h > 0$, choose $T > 0$, set $C_T = Q_R(T)/T^4$, and define

$$t_* = \min\{T, (1-\rho)h/(2C_T)\}, \qquad c = (1+\rho)/2. \tag{6}$$

For $0 < t \le t_*$, $\Delta_n(t) \le cht^3 < ht^3 \le \mathcal{B}(t)$.

First discrepancy. Signature and decoded series F, G agreeing through degree two have $D_3 = F_3 - G_3 = (\log_\otimes F)_3 - (\log_\otimes G)_3$, a cubic Lie tensor. Cubic gauge and uniform response tail give first bound in (5); degree-three certificate term gives second. Thus

$$\left|\frac{\Delta_n(t)}{\mathcal{B}(t)} - \rho\right| \le \frac{Q_R(t) + \rho Q_H(t)}{ht^3}.$$

Both tails are $O(t^4)$, independently of n. Interval factors first differ by $\ell_{i,3}$. Nonconstant factors raise degree, so $D_3 = \sum_i \ell_{i,3}$.

Full series control. BV factorial bound $\|S_j(X)\|_{\mathrm{HS}} \le L^j/j!$ majorizes F by e^{Lt}. Tensor products majorize G_π by e^{at+bt^2}. Area gauge majorizes matrix decoder by $e^{at+(b/2)t^2}$. Removing degrees zero to three gives Q_R, Q_H. Equivalently,

$$T_4(a,b;t) = e^p - P_3(p) + q(e^p - P_1(p)) + e^p(e^q - P_1(q)), \quad p = at,\ q = bt^2.$$

Nonnegative coefficients imply $Q_R(t) \le C_T t^4$ on $0 \le t \le T$, which proves (6) and the strict gap.

BV stability. For $X_0 = Y_0 = 0$, variations at most L, and $\eta = \|X - Y\|_{1\text{-var}}$, interpolation and insertion of one $d(Y - X)$ into iterated integrals give

$$\|S_j(X) - S_j(Y)\|_{\mathrm{HS}} \le \frac{\eta L^{j-1}}{(j-1)!}.$$

At insertion time u, let $L_{\mathrm{before}}, L_{\mathrm{after}}$ be variations of interpolated path before and after u. Summing positions gives $(L_{\mathrm{before}} + L_{\mathrm{after}})^{j-1}/(j-1)! \le L^{j-1}/(j-1)!$. Integrate against variation measure of $Y - X$, then interpolation parameter. In

$$\ell_3 = S_3 - \tfrac{1}{2}(S_1 \otimes S_2 + S_2 \otimes S_1) + \tfrac{1}{3}S_1^{\otimes 3},$$

with $\ell_3(X) = (\log_{\otimes} S(X))_3$, the three differences have bounds $\eta L^2/2$, $3\eta L^2/2$, and ηL^2. Therefore $\|\ell_3(X) - \ell_3(Y)\|_{\mathrm{HS}} \le 3\eta L^2$. On a common partition, local difference variations sum to η, giving the same bound for D_3. With $L_i = \|X_i\|_{1\text{-var}}$, $a \le L$ and $b \le \sum_i L_i^2 \le L^2$. For baseline X, let $L_X = \|X\|_{1\text{-var}}$, $h_X = \|D_3(X)\|_{\mathrm{HS}} > 0$, and set

$$L = L_X + 1, \qquad \eta_* = \min\{1, h_X/(6L^2)\}.$$

Paths Y with $Y_0 = 0$ and $\|X - Y\|_{1\text{-var}} \le \eta_*$ have cubic norm at least $h_X/2$. Replacing Q_R by $Q_R^*(t) = T_4(L, 0; t) + T_4(L, L^2/2; t)$ and h by $h_X/2$ in (6) gives a common positive radius.

Mixed blocks. For $v_1, v_2 \in \mathbb{R}^2$ and $\alpha_1, \alpha_2 \in \mathbb{R}$, compare

$$e_{\otimes}^{tv_1 + t^2\alpha_1 Z} e_{\otimes}^{tv_2 + t^2\alpha_2 Z} \quad \text{with} \quad e_{\otimes}^{t(v_1+v_2) + t^2(\alpha_1+\alpha_2+\omega/2)Z}, \qquad \omega = \det(v_1, v_2).$$

Baker–Campbell–Hausdorff expansion gives cubic discrepancy $t^3[w, Z]$, where

$$w = \tfrac{1}{2}(\alpha_2 v_1 - \alpha_1 v_2) + \tfrac{\omega}{12}(v_1 - v_2).$$

Cubic gauge and tail argument apply when $w \neq 0$, with response tail $T_4(a_f, b_f/2; t) + T_4(a_c, b_c/2; t)$, where

$$a_f = \|v_1\|_2 + \|v_2\|_2, \quad b_f = \sqrt{2}(|\alpha_1| + |\alpha_2|), \quad a_c = \|v_1 + v_2\|_2, \quad b_c = \sqrt{2}|\alpha_1 + \alpha_2 + \omega/2|.$$

For independent v_1, v_2, $w = 0$ exactly when $\alpha_1 = \alpha_2 = -\omega/6$. Pure planar area blocks commute.

B Gram certificates and all-scale chain proof

Let blocks $U_1, \ldots, U_\ell$ have common tensor degree $k \ge 1$ and Gram matrix $g_{ij} = \langle U_i, U_j \rangle_{\mathrm{HS}}$. For nonnegative integer tuple $\alpha = (\alpha_1, \ldots, \alpha_\ell)$, let $|\alpha| = \sum_i \alpha_i$, $\alpha! = \prod_i \alpha_i!$, and $\binom{m}{\alpha} = m!/\alpha!$ when $|\alpha| = m$. Define $w(\alpha)$ as the nondecreasing word containing α_i copies of index i, and $W_\alpha = U_1^{\otimes\alpha_1} \otimes \cdots \otimes U_\ell^{\otimes\alpha_\ell}$. Write F_m, G_m for coefficients of λ^m in ordered and merged series; they have tensor degree km:

$$F_m = \sum_{|\alpha|=m} W_\alpha/\alpha!, \qquad G_m = (\sum_i U_i)^{\otimes m}/m!.$$

With $H_m = m!\,\|F_m - G_m\|_{\mathrm{HS}}$, tensor inner products give

$$H_m^2 = \sum_{|\alpha|=|\beta|=m} \binom{m}{\alpha}\binom{m}{\beta} \prod_{v=1}^{m} g_{w(\alpha)_v, w(\beta)_v} - \left(\sum_{i,j} g_{ij}\right)^m. \tag{7}$$

Here $\langle W_\alpha, W_\beta \rangle_{\mathrm{HS}}$ is the displayed product. Writing $s_i = \sum_j g_{ij}$, the multinomial theorem gives $\langle F_m, G_m \rangle = (\sum_i s_i)^m/(m!)^2 = \|G_m\|_{\mathrm{HS}}^2$, which proves (7).

Response bound. For $X_i = R_A(U_i)$, put

$$M = \sup_{|A|\le 1} \sum_i \|X_i\|_{\mathrm{op}}, \qquad K = \sup_{|A|\le 1} \sum_{i<j} \|[X_i, X_j]\|_{\mathrm{op}}.$$

If $M = 0$, response error vanishes and positive chain certificate gives separation. For $M > 0$, two-factor commutator representation [7] gives

$$\left\| e^{\lambda X} e^{\lambda Y} - e^{\lambda(X+Y)} \right\|_{\mathrm{op}} \le \frac{\lambda^2}{2} \|[X, Y]\|_{\mathrm{op}} e^{\lambda(\|X\|_{\mathrm{op}} + \|Y\|_{\mathrm{op}})}.$$

For completeness, differentiate $e^{(\lambda-s)(X+Y)} e^{sX} e^{sY}$ and use

$$[e^{sX}, Y] = \int_0^s e^{(s-u)X} [X, Y] e^{uX}\, du.$$

Induction over factors, with $[\sum_{i<j} X_i, X_j] = \sum_{i<j} [X_i, X_j]$, yields

$$\Delta_n(\lambda) \le \frac{\lambda^2 K}{2} e^{\lambda M}.$$

Comparing power coefficients with $\mathcal{B}(\lambda) = \sum_{m\ge 2} \lambda^m H_m/m!$, a sufficient condition is

$$K < \inf_{m\ge 2} \frac{2H_m}{m(m-1)M^{m-2}} \quad (M > 0).$$

Analytic verification for chains. For $U_i = [e_i, e_{i+1}]$, blocks are mutually orthogonal with norm $\sqrt{2}$. Let $a_i = \|A_i\|_F$ and $c_\ell = \cos(\pi/(\ell+2))$. The path-graph largest eigenvalue gives $\sum_i a_i a_{i+1} \le c_\ell \sum_i a_i^2 \le c_\ell$. Böttcher–Wenzel therefore gives

$$M \le \sqrt{2} c_\ell < \sqrt{2}, \qquad K \le 2\sqrt{2} \sum_{i<j} (a_i a_{i+1})(a_j a_{j+1}) \le \sqrt{2} c_\ell^2 < \sqrt{2}.$$

These inequalities hold for each tuple before taking suprema. Retain only nonmonotone words over the first three orthogonal blocks. There are $3^m - \binom{m+2}{2}$ such words, each absent from F_m and having coefficient $1/m!$ in G_m. Hence

$$H_m^2 \ge 2^m (3^m - \binom{m+2}{2}) \ge 2^m 3^{m-1}, \qquad m \ge 2.$$

The counting inequality follows at $m = 2$ and by induction: $\binom{m+3}{2}/\binom{m+2}{2} = (m+3)/(m+1) < 3$. Consequently every coefficient threshold is at least

$$L_m = \frac{4\, 3^{(m-1)/2}}{m(m-1)}.$$

Its consecutive ratio is $\sqrt{3}(m-1)/(m+1)$: below one up to $m = 3$, and above one for $m \ge 4$. For $m = 4 + s$, the latter follows from $3(m-1)^2 - (m+1)^2 = 2s^2 + 8s + 2 > 0$. Thus $\inf_{m\ge 2} L_m = L_4 = \sqrt{3} > \sqrt{2} > K$. Summing $H_m \ge (\sqrt{6})^m/\sqrt{3}$ gives the last bound in (3); the same coefficient comparison gives its middle strict inequality, proving Theorem 2.

C Euler minimax proof

Put $x_i = |A| \, \|X_i\|_{1\text{-var}}$ and $R = \sum_i x_i$. Expanding all exact interval factors, the Euler product retains precisely multi-indices with every interval degree at most N. Factorial estimates give

$$\left\| \Phi^A(X) - E^A_{N,\pi}(X) \right\|_{\mathrm{op}} \leq \sum_{\max j_i > N} \prod_i \frac{x_i^{j_i}}{j_i!} = e^R - \prod_i P_N(x_i).$$

The majorant increases in each x_i. For $N \geq 2$, direct coefficient comparison gives

$$P_{N-1}(x)^2 - P_N(x) P_{N-2}(x) = \frac{x^{N-1}}{N!} \sum_{j=0}^{N-1} (N-j) \frac{x^j}{j!} \geq 0.$$

For $N = 1$, $\log P_1 = \log(1+x)$ is concave as well. Thus $\log P_N$ is concave, and Jensen maximizes the product at equal $x_i = R/m$. An equal-variation partition is independent of A and gives the uniform bound $\mathcal{E}(N, m, r)$ on the normalized path class.

For $X_t = te_1$, choose rank-one orthogonal projection Q, $A_1 = rQ$, and other matrices zero. On its range, with $\Delta t_i = t_i - t_{i-1}$, error is $e^r - \prod_i P_N(r\Delta t_i)$. Jensen gives error at least $\mathcal{E}(N, m, r)$ for every partition; equal intervals attain it, proving (4).

D Storage limit arguments

Witt's formula gives generic log-signature dimension:

$$\Lambda_d(N) = \sum_{j=1}^{N} \frac{1}{j} \sum_{v \mid j} \mu(v) d^{j/v},$$

where $\mu(1) = 1$; $\mu(v) = (-1)^s$ for a product of s distinct primes, and $\mu(v) = 0$ when v has a squared prime divisor.

Limiting storage statement. Let $C_{d,k}(z, \varepsilon)$ be minimum $m\Lambda_d(kq)$ over positive integers q, m satisfying $\mathcal{E}(q, m, z) \leq \varepsilon$, with $d \geq 2, k \geq 1$ fixed and $z, \varepsilon > 0$. Stored depth is kq. Put $\beta = k \log d$, $\lambda = z + \log(z/\varepsilon)$, with natural logs. For any optimizing pair (q_*, m_*), in either limit $z \to \infty$ at fixed $\varepsilon > 0$, or $\varepsilon \downarrow 0$ at fixed $z > 0$,

$$q_* \sim \sqrt{\lambda/\beta}, \qquad \log(m_*/z) \sim \sqrt{\beta\lambda}, \qquad \log(C_{d,k}/z) \sim 2\sqrt{\beta\lambda}. \tag{8}$$

Here $f \sim g$ means $f/g \to 1$. Replacing constraint $\mathcal{E}(q, m, z) \leq \varepsilon$ by $B(q, m, z) \leq \varepsilon$ changes minimum cost by a factor tending to one. For $k = 1$, leading BV rates agree with the (ε, r)-representation paper.

Proof. Write $x = z/m$, $a = e^{-x} \sum_{j>q} x^j/j!$, and $B = e^z z^{q+1}/((q+1)!m^q)$. Repeated integration of e^{-u} gives

$$e^{-x} \frac{x^{q+1}}{(q+1)!} \leq a \leq \frac{x^{q+1}}{(q+1)!}.$$

Since $\mathcal{E} = e^z[1 - (1-a)^m]$, Bernoulli's inequality yields

$$\frac{e^{-x}}{1+ma} \leq \frac{\mathcal{E}}{B} \leq 1. \tag{9}$$

Feasibility $\mathcal{E} \leq \varepsilon < e^z$ and $1 - (1-a)^m \geq 1 - e^{-ma}$ also imply

$$ma \leq -\log(1 - \varepsilon e^{-z}). \tag{10}$$

Optimizer regime. For fixed d, k, Witt's formula gives $\log \Lambda_d(kq) = \beta q - \log q + O(1)$. Take q nearest $\sqrt{\lambda/\beta}$ and

$$m = \left\lceil z \exp\left(\frac{\lambda - \log((q+1)!)}{q}\right)\right\rceil.$$

This pair is B-feasible. Stirling and relative rounding error give $\log(C_{d,k}/z) \le 2\sqrt{\beta\lambda} + O(\log\lambda)$. Since $m \ge 1$, every optimizer has $q = O(\sqrt{\lambda})$; for large z, use $\log z = O(\log\lambda)$.

At fixed z, bounded m and $q = O(\sqrt{\lambda})$ would, by the first omitted term and $1-(1-a)^m \ge a$, give $\log\mathcal{E} \ge -O(\sqrt{\lambda}\log\lambda)$, contradicting $\log\varepsilon = -\lambda + O(1)$. Thus $m \to \infty$ and $x \to 0$. At $z \to \infty$, $m = 1$ gives $\log P_q(z) = O(\sqrt{z}\log z) = o(z)$, so $\mathcal{E} \sim e^z$, infeasible. If $m \ge 2$ and $x \ge c > 0$, then

$$\mathcal{E} \ge e^{z-x}\frac{x^{q+1}}{(q+1)!} \quad\Longrightarrow\quad \log\mathcal{E} \ge z/2 - O(\sqrt{z}\log z),$$

also infeasible. Hence $x \to 0$, and $m \to \infty$, in both regimes.

At fixed z, $ma \le zx^q/(q+1)! \le zx/2 \to 0$ once $x < 1$. For large z, (10) gives $ma \to 0$. It follows from (9) that $\mathcal{E}/B \to 1$ at every optimizer, so $\log B \le \log\varepsilon + o(1)$. Consequently

$$\log(C_{d,k}/z) \ge \beta q + \lambda/q - O(\log\lambda).$$

The minimum $2\sqrt{\beta\lambda}$ forces $q/\sqrt{\lambda/\beta} \to 1$ by the preceding upper and lower bounds. Subtracting $\log\Lambda_d(kq)$ gives the formula for $\log(m/z)$.

Finally, at an exact optimizer inflate m by $\max\{1, (B/\varepsilon)^{1/q}\}$, rounding upward. This is B-feasible and costs a relative $1 + o(1)$ since $B/\varepsilon \le 1 + o(1)$ and $m \to \infty$. Since $\mathcal{E} \le B$, this proves convergence of minimum-cost ratio to one.

E Engel gauge and structured storage scope

Fix $d, n \ge 2$ and $k \ge 2$. Define $\mathrm{ad}_U(V) = [U, V]$ and Engel direction $E_k = \mathrm{ad}_{e_1}^{k-1}(e_2)$. Put

$$h_k = \sqrt{\binom{2k-2}{k-1}}, \quad g_k = [2(k-1)]^{(k-1)/2}k^{-k/2}, \quad s_k = g_k/\sqrt{2}.$$

Here $h_k = \|E_k\|_{\mathrm{HS}}$, $g_k = \gamma_{k,n}(E_k)$, and s_k is an attained positive eigenvalue. For amplitude $b > 0$, pure degree-k weakly geometric rough path has signature increments $S_{u,v} = \exp_\otimes((v-u)bE_k)$, $0 \le u \le v \le 1$, and roughness exponent $p \ge k$. Absorb coefficient radius r into $a = br^k$. Let $C_k^{\mathrm{true}}(a,\varepsilon)$ minimize generic Euler cost $m\Lambda_d(N)$ over $N, m \ge 1$ meeting response tolerance $\varepsilon > 0$, with path-dependent, coefficient-independent partitions. When $\varepsilon < e^{as_k} - 1$, true Euler storage is bracketed by

$$C_{d,k}(as_k,\varepsilon) \le C_k^{\mathrm{true}}(a,\varepsilon) \le C_{d,k}(ag_k,\varepsilon) \le C_{d,k}(ah_k,\varepsilon).$$

Consequently, at fixed d, k, ε,

$$\liminf_{a\to\infty}\frac{1}{\sqrt{a}}\log\frac{C_{d,k}(ah_k,\varepsilon)}{C_k^{\mathrm{true}}(a,\varepsilon)} \ge 2\sqrt{k\log d}\,(\sqrt{h_k} - \sqrt{g_k}) > 0.$$

A single depth-k log-ODE factor represents this path exactly.

Gauges and storage proof. Repeated Böttcher–Wenzel gives

$$\left\|\mathrm{ad}_A^{k-1}(B)\right\|_{\mathrm{op}} \le 2^{(k-1)/2}\,\|A\|_F^{k-1}\,\|B\|_F.$$

Maximizing the upper bound under $\|A\|_F^2 + \|B\|_F^2 \le 1$ gives squared norms $(k-1)/k, 1/k$ and value g_k. A diagonal block $A = \sqrt{(k-1)/(2k)}\,\mathrm{diag}(1,-1)$ and $B = E_{12}/\sqrt{k}$ attain operator norm g_k. Binomial expansion of E_k gives HS norm h_k. Stirling gives $g_k/h_k = O(k^{-1/4}2^{-k/2})$.

At effective amplitude a, let $\delta_k(q,m,a)$ be worst unit-budget response error at Euler depth kq on equal m-interval partition, $q, m \ge 1$. Nonnegative scalar error coefficients give

$$\mathcal{E}(q,m,as_k) \le \delta_k(q,m,a) \le \mathcal{E}(q,m,ag_k) < \mathcal{E}(q,m,ah_k).$$

For the lower witness, set $\alpha = \sqrt{(k-1)/k}$, $\eta = 1/\sqrt{k}$:

$$A_1 = \frac{\alpha}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad A_2 = \frac{\eta}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ (-1)^{k-1} & 0 \end{pmatrix}.$$

Embedded in dimension n, this unit tuple evaluates E_k to a symmetric block with off-diagonal entries s_k, hence eigenvalue s_k. Jensen gives the same lower bound for adaptive partitions. Let $c_j \ge 0$ be coefficients of $e^x - P_q(x/m)^m$. Each tensor-degree-kj error is $c_j a^j E_k^{\otimes j}$. Its norm is $c_j(ah_k)^j$, whose sum equals $\mathcal{E}(q,m,ah_k)$.

For $\varepsilon < e^{as_k} - 1$, depths below k are infeasible. Depth $N \ge k$ gives the same polynomial as $k\lfloor N/k \rfloor$, with no smaller storage. This proves storage sandwich; (8) gives amplitude bound. At fixed a, all bracket endpoints have common leading logarithmic rate $2\sqrt{k \log d \log(1/\varepsilon)}$.

F Rational certificates and reproducibility

For the one-interval corner of Section 2, with $Z = [e_1, e_2]$,

$$D_3 = ([e_1, Z] - [e_2, Z])/12, \qquad h = \sqrt{3}/6, \quad a = \sqrt{2}, \quad b = \sqrt{2}/2.$$

For rational $x \ge 0$ and integer $J \ge 0$ with $x < J + 2$,

$$P_J(x) \le e^x \le P_J(x) + \frac{x^{J+1}}{(J+1)!}\frac{1}{1 - x/(J+2)}.$$

Successive omitted-term ratios are bounded by $x/(J+2)$. Displayed example enclosures use $J = 24$. Integer squares verify

$$\sqrt{2} < 707107/500000, \quad \sqrt{3} > 34641/20000, \quad \sqrt{6} < 244949/100000.$$

For the corner, $\rho h = \sqrt{6}/27$. Positive coefficients permit replacing $\sqrt{2}$ parameters by their upper bounds in Q_R; apply the exponential enclosure, and use the lower $\sqrt{3}$ bound in ht^3. Exact fractions then give (2).

For the storage example, bound $e = e^1$ by the same rational enclosure. For $d = 2$, first five cumulative dimensions are $2, 3, 5, 8, 14$. Enumerating every (N, m) of cost at most 8 for $\mathcal{E}$, and at most 14 for B, gives unique optima in the table. Higher depths cost more with $m = 1$. Each feasibility inequality is strictly decided by the rational bounds.

Accompanying source archive contains proofs, modules, tests, runners, notebook and curated evidence. Build runner regenerates grid statistics from retained CSV and rechecks every optimum and tie set. General floating-point searches are uncertified.

References

[1] Ferrucci, Emilio; Perrée, Oliver; Lyons, Terry. *Concise (ε, r)-representations of a path*. 2026. https://arxiv.org/abs/2607.26281.

[2] Böttcher, Albrecht; Wenzel, David. *The Frobenius norm and the commutator*. Linear Algebra and its Applications 429, 1864–1885 (2008). https://doi.org/10.1016/j.laa.2008.05.020.

[3] Chevyrev, Ilya; Lyons, Terry. *Characteristic functions of measures on geometric rough paths*. The Annals of Probability 44, 4049–4082 (2016). https://arxiv.org/abs/1307.3580.

[4] Nikolaev, Rusalin S. *Identities of two variables in the Lie algebra sl(2, K) over a field of characteristic zero*. Pliska Studia Mathematica Bulgarica 8, 65–76 (1986). https://www.math.bas.bg/pliska/Pliska-08/Pliska-08-1986-065-076.pdf.

[5] Boedihardjo, Horatio; Geng, Xi; Souris, Nikolaos P. *Path developments and tail asymptotics of signature for pure rough paths*. Advances in Mathematics 364, 107043 (2020). https://arxiv.org/abs/1811.12170.

[6] Li, Nannan; Gao, Xing. *Exact signature tail asymptotics for pure rough paths*. 2026. https://arxiv.org/abs/2606.30055.

[7] Iserles, Arieh; Kropielnicka, Karolina. *An elementary approach to splittings of unbounded operators*. 2024. https://arxiv.org/abs/2401.06635.

[8] Bayer, Christian; Breneis, Simon; Lyons, Terry. *An adaptive algorithm for rough differential equations*. 2023. https://arxiv.org/abs/2307.12590.